ICP Skill Factory — Skill Candidate Inventory (Phase 2)

Version: R5 (2026-03-13) **Total Candidates:** 173 pre-normalization / 171 post-normalization **Domains Covered:** 14 (D01–D14) **Source:** Extracted from ID01–ID14, ID21–ID25, GOV-SD, GOV-SDP, and practical engineering knowledge (PRAC)

Legend

- **Source**: ID01–ID14 = document source (Tier 2 exemplar); ID21–ID25 = org procedures (Tier 3 contextual); GOV-SD/GOV-SDP = governance framework (Tier 1 authoritative); PRAC = practical engineering knowledge
- **Confidence**: H = High (directly stated in source), H+ = High with deliverable exemplar + governance confirmation, M = Medium (strongly implied), L = Low (inferred from context)
- **Type**: ANA=Analysis, DES=Design, ENG=Engineering, TST=Testing, DOC=Documentation, MGT=Management, VER=Verification, GOV=Governance, INT=Integration, OPS=Operations
- **R5 markers**: ▲ = new in R5 from governance (Tier 1); △ = new in R5 from exemplars (Tier 2/3)

D01 — OT Cybersecurity (36 candidates; +8 in R5)

SC ID	Skill Name (EN)	技能名稱 (ZH)	Subdomain	Type	Lifecycle	Source	Confidence
SC-D01-001	Zone/Conduit Architecture Design	Zone/Conduit 架構設計	D01.1	DES	R1,R2	ID01 §7.4.1.2; ID02 A.9 §10.2	H
SC-D01-002	Defense-in-Depth Strategy Design	縱深防禦策略設計	D01.1	DES	R1,R2	ID01 §6.5.1.3; ID02 A.9 §2	H
SC-D01-003	Firewall Rule Planning	防火牆規則規劃	D01.1	ENG	R2,R3	ID02 A.9 §10.1	H
SC-D01-004	Network Segmentation Documentation	網路分段文件	D01.1	DOC	R2,R3	ID01 §7.4.1.2; ID02 C.3	H
SC-D01-005	Asset Inventory Development	資產清冊建立	D01.2	ANA	R0,R1	ID01 §7.2.1; ID02 A.6; ID02 C.4	H
SC-D01-006	Threat and Risk Assessment (Preliminary)	初步威脅風險評估	D01.2	ANA	R1	ID01 §7.2; ID02 A.8	H
SC-D01-007	Detailed Risk Assessment	詳細風險評估	D01.2	ANA	R2	ID01 §7.3; ID03 §5.6.4	H

SC ID	Skill Name (EN)	技能名稱 (ZH)	Subdomain	Type	Lifecycle	Source	Confidence
SC-D01-008	STRIDE/DREAD Threat Modeling	STRIDE/DREAD 威脅建模	D01.2	ANA	R1,R2	ID01 §5.0 ref[32]; PRAC	M
SC-D01-009	Risk Classification Matrix Development	風險分類矩陣建立	D01.2	ANA	R1	ID03 §5.4.1	H
SC-D01-010	Security Level Target (SL-T) Assessment	安全等級目標評估	D01.3	ANA	R1	ID01 §6.5.1.2; ID03 §5.4.2	H
SC-D01-011	IEC 62443 Compliance Gap Analysis	IEC 62443 合規差距分析	D01.3	ANA	R1,R4	ID01 §6.6.3; ID02 C.1-C.8	H
SC-D01-012	Security Audit Execution	安全稽核執行	D01.3	VER	R4	ID01 §7.8.3.1	H
SC-D01-013	Gate Review Preparation and Execution	閘門審查準備與執行	D01.3	GOV	R0-R5	ID01 §6.5.1.1.3, §6.5.1.2.4	H
SC-D01-014	SIEM Configuration and Tuning	SIEM 配置與調校	D01.4	ENG	R3,R4	ID01 §7.8.4.2; ID02 A.4 §9	H
SC-D01-015	Security Alarm Rule Design	安全告警規則設計	D01.4	DES	R3	ID01 §7.8.4.1	H
SC-D01-016	Incident Response Procedure Development	事件回應程序撰寫	D01.4	DOC	R3,R4	ID01 §7.8.5; ID03 §5.5.2	H
SC-D01-017	Security Incident Investigation and Forensics	安全事件調查與鑑識	D01.4	ANA	R4	ID01 §6.5.2.5	M
SC-D01-018	Continuous Security Monitoring	持續安全監控	D01.4	OPS	R4	ID01 §6.5.2.6; §7.8.4.2	H
SC-D01-019	Endpoint Hardening Implementation	端點安全加固實施	D01.5	ENG	R3	ID01 §7.4.1.5; ID02 A.10	H
SC-D01-020	Account and Access Control Management	帳號與存取控制管理	D01.5	ENG	R3,R4	ID01 §7.4.1.3; ID02 A.4 §4	H

SC ID	Skill Name (EN)	技能名稱 (ZH)	Subdomain	Type	Lifecycle	Source	Confidence
SC-D01-021	Security Patch Management	安全補丁管理	D01.5	OPS	R3,R4	ID01 §7.4.1.11; ID02 A.4 §8.5	H
SC-D01-022	Backup and Restore Procedure Design	備份與還原程序設計	D01.5	DES	R3	ID01 §7.4.1.12; ID02 A.4 §6	H
SC-D01-023	Malware Protection Implementation	惡意程式防護實施	D01.5	ENG	R3	ID01 §7.4.1.10; ID02 A.4 §7	H
SC-D01-024	Vendor Security Risk Assessment	供應商安全風險評估	D01.6	ANA	R0,R1	ID01 §7.1.1.2; ID02 A.2	H
SC-D01-025	SBOM Analysis and Management	SBOM 分析與管理	D01.6	ANA	R3,R4	ID03 §5.3.1; PRAC	M
SC-D01-026	Third-Party Component Security Verification	第三方元件安全驗證	D01.6	VER	R2,R3	ID01 §6.4; ID02 B.3	M
SC-D01-027	SIS Security Control Implementation	SIS 安全控制實施	D01.7★	ENG	R2,R3	ID01 §7.4.1.7; ID02 A.9 §12	H
SC-D01-028	Remote Access Security Configuration	遠端存取安全配置	D01.5	ENG	R3	ID01 §7.4.1.8; ID02 A.9 §9	H
SC-D01-029 △	Security Management Plan Development	安全管理計畫撰寫	D01.3	DOC	R0,R1	ID04 (Tier 2 format example)	H
SC-D01-030 △	Security Policies and Procedures Plan Development	安全政策與程序計畫撰寫	D01.5	DOC	R1,R2	ID07 (Tier 2); ID01 §7.1.1.4	H
SC-D01-031 △	Vendor Security Management Plan Development	供應商安全管理計畫撰寫	D01.6	DOC	R0,R1	ID05 (Tier 2); ID01 §7.1.1.2	H

SC ID	Skill Name (EN)	技能名稱 (ZH)	Subdomain	Type	Lifecycle	Source	Confidence
SC-D01-032 △	Threat Intelligence Collection and Analysis	威脅情資蒐集與分析	D01.4	ANA	R4	ID24 §5.3–5.4 (Tier 3)	H
SC-D01-033 △	Data Classification Policy Development	資料分類政策制定	D01.5	GOV	R1	ID23 (Tier 3); GOV-SD BOUNDARY-008	H
SC-D01-034 △	Security Solution Integration Plan Development	安全解決方案整合計畫撰寫	D01.5	DOC	R3	ID04 §5.0 (Tier 2)	M
SC-D01-035 ▲	Integrated Risk Assessment Execution (IEC+FMEA+HAZOP)	整合風險評估執行	D01.2	ANA	R1,R2	GOV-SD (triple-method mandatory)	H
SC-D01-036 ▲	Risk Source Traceability & Residual Risk Register	風險溯源與殘餘風險登錄	D01.2	ANA	R1–R3	GOV-SD (T-XXX/FM-XXX/HAZ-XXX; Gate 3 20% sampling)	H

D02 — System Architecture (12 candidates)

SC ID	Skill Name (EN)	技能名稱 (ZH)	Subdomain	Type	Lifecycle	Source	Confidence
SC-D02-001	OT Network Topology Design	OT 網路拓撲設計	D02.1	DES	R1,R2	ID02 A.7; ID01 §7.4.1.2	H
SC-D02-002	Network Redundancy Design (RSTP/Ring)	網路冗餘設計	D02.1	DES	R2	PRAC	M
SC-D02-003	Interface Control Document (ICD) Development	介面控制文件撰寫	D02.2	DOC	R2	PRAC	M
SC-D02-004	Data Flow Diagram Development	資料流圖繪製	D02.2	DOC	R1,R2	ID02 A.5; ID01 §7.2.1	H

SC ID	Skill Name (EN)	技能名稱 (ZH)	Subdomain	Type	Lifecycle	Source	Confidence
SC-D02-005	Industrial Protocol Architecture Design	工業協定架構設計	D02.3	DES	R1,R2	PRAC	M
SC-D02-006	High-Availability Architecture Design	高可用架構設計	D02.4	DES	R2	PRAC	M
SC-D02-007	RTO/RPO Planning	RTO/RPO 規劃	D02.4	DES	R2	PRAC	M
SC-D02-008	Edge Computing Deployment Design	邊緣計算部署設計	D02.5	DES	R2	PRAC	L
SC-D02-009	Architecture Decision Record (ADR) Writing	架構決策記錄撰寫	D02.6	DOC	R1,R2	PRAC	M
SC-D02-010	Technology Selection Evaluation	技術選型評估	D02.6	ANA	R0,R1	PRAC	M
SC-D02-011	Simple Network Diagram Development	簡易網路圖繪製	D02.1	DOC	R1	ID02 A.7; ID03 Table 2 Doc 1.05	H
SC-D02-012	Architecture Review Facilitation	架構審查主持	D02.6	GOV	R2	ID01 §7.3.2	H

D03 — Power System Engineering (10 candidates)

SC ID	Skill Name (EN)	技能名稱 (ZH)	Subdomain	Type	Lifecycle	Source	Confidence
SC-D03-001	Power Flow Analysis	潮流分析	D03.1	ANA	R1,R2	PRAC	H
SC-D03-002	Short Circuit Current Analysis	短路電流分析	D03.1	ANA	R1,R2	PRAC	H
SC-D03-003	Voltage Stability Assessment	電壓穩定度評估	D03.1	ANA	R2	PRAC	M

SC ID	Skill Name (EN)	技能名稱 (ZH)	Subdomain	Type	Lifecycle	Source	Confidence
SC-D03-004	PV System Grid Integration Design	PV 系統併網設計	D03.2	DES	R1,R2	PRAC	M
SC-D03-005	BESS Capacity Planning	BESS 容量規劃	D03.3	DES	R1,R2	PRAC	M
SC-D03-006	VPP Dispatch Algorithm Design	VPP 調度演算法設計	D03.4	DES	R2	PRAC	M
SC-D03-007	DER Aggregation Strategy Design	DER 聚合策略設計	D03.4	DES	R1,R2	PRAC	M
SC-D03-008	Harmonic Analysis and Filter Design	諧波分析與濾波器設計	D03.5	ANA	R2	PRAC	M
SC-D03-009	Transient Stability Simulation	暫態穩定度模擬	D03.6	ANA	R2	PRAC	M
SC-D03-010	Power System Modeling (ETAP/PSS/E)	電力系統建模	D03.6	ENG	R2	PRAC	M

D04 — Protection Engineering (6 candidates)

SC ID	Skill Name (EN)	技能名稱 (ZH)	Subdomain	Type	Lifecycle	Source	Confidence
SC-D04-001	Overcurrent Protection Coordination	過電流保護協調	D04.1	DES	R2	PRAC	H
SC-D04-002	Distance Protection Setting Calculation	距離保護整定計算	D04.1	ANA	R2	PRAC	M
SC-D04-003	Relay Selection and Parameter Setting	繼電器選型與參數設定	D04.2	ENG	R2,R3	PRAC	H
SC-D04-004	Protection Logic Diagram Development	保護邏輯圖繪製	D04.3	DES	R2	PRAC	M

SC ID	Skill Name (EN)	技能名稱 (ZH)	Subdomain	Type	Lifecycle	Source	Confidence
SC-D04-005	Fault Recording Analysis	故障錄波分析	D04.4	ANA	R4	PRAC	M
SC-D04-006	Protection Relay Testing	保護繼電器測試	D04.2	TST	R3	PRAC	H

D05 — Control System Engineering (14 candidates)

SC ID	Skill Name (EN)	技能名稱 (ZH)	Subdomain	Type	Lifecycle	Source	Confidence
SC-D05-001	SCADA Point List Development	SCADA 點位清單建立	D05.1	ENG	R2	PRAC	H
SC-D05-002	SCADA Database Structure Design	SCADA 資料庫結構設計	D05.1	DES	R2	PRAC	H
SC-D05-003	EMS AGC Configuration	EMS AGC 配置	D05.2	ENG	R2,R3	PRAC	M
SC-D05-004	DERMS DER Management Strategy Setting	DERMS DER 管理策略設定	D05.2	ENG	R2,R3	PRAC	M
SC-D05-005	HMI Screen Design	HMI 畫面設計	D05.3	DES	R2,R3	PRAC	H
SC-D05-006	Alarm Hierarchy and Configuration Design	告警層級配置設計	D05.3	DES	R2,R3	PRAC	H
SC-D05-007	PLC Ladder Logic Programming	PLC 階梯圖程式撰寫	D05.4	ENG	R2,R3	PRAC	H
SC-D05-008	Structured Text Programming	結構化文字程式開發	D05.4	ENG	R2,R3	PRAC	M
SC-D05-009	Modbus Mapping Configuration	Modbus 映射配置	D05.5	ENG	R3	PRAC	H

SC ID	Skill Name (EN)	技能名稱 (ZH)	Subdomain	Type	Lifecycle	Source	Confidence
SC-D05-010	IEC 61850 SCL Configuration	IEC 61850 SCL 配置	D05.5	ENG	R3	PRAC	M
SC-D05-011	OPC UA Server/Client Configuration	OPC UA 伺服器/客戶端配置	D05.5	ENG	R3	PRAC	M
SC-D05-012	PID Control Tuning	PID 控制調參	D05.6	ENG	R3	PRAC	M
SC-D05-013	Load Management Strategy Design	負載管理策略設計	D05.6	DES	R2	PRAC	M
SC-D05-014	Frequency Regulation Control Design	頻率調節控制設計	D05.6	DES	R2	PRAC	M

D06 — Panel Engineering (6 candidates)

SC ID	Skill Name (EN)	技能名稱 (ZH)	Subdomain	Type	Lifecycle	Source	Confidence
SC-D06-001	Panel Layout Design and Thermal Calculation	盤面佈局設計與散熱計算	D06.1	DES	R2	PRAC	H
SC-D06-002	Wiring Diagram Development	配線圖繪製	D06.2	DES	R2	PRAC	H
SC-D06-003	Terminal Block Schedule Development	端子排表建立	D06.2	DOC	R2	PRAC	H
SC-D06-004	Fabrication Drawing Production (CAD)	施工圖 CAD 出圖	D06.3	ENG	R2	PRAC	H
SC-D06-005	Wire Sizing Calculation	線徑選用計算	D06.2	ANA	R2	PRAC	M
SC-D06-006	Component Selection and Specification Writing	元件選型與規範書撰寫	D06.4	DES	R1,R2	PRAC	H

D07 — Integration Engineering (7 candidates)

SC ID	Skill Name (EN)	技能名稱 (ZH)	Subdomain	Type	Lifecycle	Source	Confidence
SC-D07-001	Interface Integration Matrix Development	介面整合矩陣建立	D07.1	DOC	R2	PRAC	M
SC-D07-002	System Integration Architecture Diagram	系統整合架構圖繪製	D07.1	DES	R2	PRAC	M
SC-D07-003	Protocol Gateway Configuration	協定閘道配置	D07.2	ENG	R3	PRAC	H
SC-D07-004	Data Format Conversion Design	資料格式轉換設計	D07.2	DES	R2,R3	PRAC	M
SC-D07-005	Cross-System Data Model Alignment	跨系統資料模型對齊	D07.3	DES	R2	PRAC	M
SC-D07-006	Timestamp Synchronization Design	時間戳同步設計	D07.3	DES	R2,R3	PRAC	M
SC-D07-007	Third-Party API Integration	第三方 API 串接	D07.4	INT	R3	PRAC	M

D08 — Testing & Commissioning (13 candidates; +2 in R5) ◆ SC-D08-012 遷移至 D10

SC ID	Skill Name (EN)	技能名稱 (ZH)	Subdomain	Type	Lifecycle	Source	Confidence
SC-D08-001	FAT Procedure Development	FAT 程序撰寫	D08.1	DOC	R3	ID01 §6.5.1.4; ID02 A.11	H
SC-D08-002	Security FAT Test Case Design	安全 FAT 測試案例設計	D08.1	TST	R3	ID02 A.11; ID03 Table 2 Doc 3.07	H
SC-D08-003	SAT Procedure Development	SAT 程序撰寫	D08.2	DOC	R3	ID02 A.11; ID03 Table 2 Doc 3.12	H
SC-D08-004	Site Acceptance Testing Execution	現場驗收測試執行	D08.2	TST	R3	ID01 §7.6; ID02 C.7	H

SC ID	Skill Name (EN)	技能名稱 (ZH)	Subdomain	Type	Lifecycle	Source	Confidence
SC-D08-005	System Security Acceptance Testing	系統安全驗收測試	D08.2	TST	R3	ID01 §6.5.1.4.3; §7.6	H
SC-D08-006	Commissioning Plan Development	試車計畫撰寫	D08.3	DOC	R3	PRAC	H
SC-D08-007	Performance Baseline Establishment	性能基線建立	D08.4	TST	R3	PRAC	M
SC-D08-008	Application Security Testing Execution	應用安全測試執行	D08.4	TST	R3	ID01 §6.5.1.4.2; §7.5	H
SC-D08-009	Penetration Testing Execution	滲透測試執行	D08.4	TST	R3	ID01 §6.5.1.4.4; ID03 Table 2 Doc 3.17	H
SC-D08-010	Vulnerability Scanning and Reporting	弱點掃描與報告	D08.4	TST	R3,R4	ID02 A.12	H
SC-D08-011	Defect Report Writing and Severity Classification	缺陷報告撰寫與分級	D08.5	DOC	R3	PRAC	H
~~SC-D08-012~~	~~System Decommissioning Execution~~	~~系統除役執行~~	~~D08.6~~	—	—	—	—
SC-D08-013 △	Site Integration Test (SIT) Protocol Development	SIT 測試協定撰寫	D08.2	DOC	R3	ID14 (Tier 2 SIT as distinct phase)	H
SC-D08-014 △	Security Inspection and Test Protocol Development	安全檢驗測試協定撰寫	D08.4	DOC	R3	ID14 (Tier 2; 14-category example)	H

D09 — Engineering Documentation (9 candidates; +1 in R5)

SC ID	Skill Name (EN)	技能名稱 (ZH)	Subdomain	Type	Lifecycle	Source	Confidence

SC ID	Skill Name (EN)	技能名稱 (ZH)	Subdomain	Type	Lifecycle	Source	Confidence
SC-D09-001	System Design Description Writing	系統設計說明書撰寫	D09.1	DOC	R2	PRAC	H
SC-D09-002	Security Functional Description Specification	安全功能描述規範撰寫	D09.1	DOC	R3	ID02 A.9; ID03 Table 2 Doc 3.02	H
SC-D09-003	Single-Line Diagram (SLD) Development	單線圖繪製	D09.2	DES	R2	PRAC	H
SC-D09-004	Document Delivery Checklist Management	文件交付清單管理	D09.4	MGT	R3	ID03 §5.5.3	H
SC-D09-005	Version Control and Archiving	版本控制與歸檔	D09.4	MGT	R0-R5	PRAC	M
SC-D09-006	Operation Manual Writing	操作手冊撰寫	D09.5	DOC	R3	PRAC	H
SC-D09-007	SOP Development	SOP 編撰	D09.5	DOC	R3,R4	PRAC	H
SC-D09-008	Training Material Development	培訓教材製作	D09.5	DOC	R3,R4	ID01 §6.3; ID03 §5.3.3	M
SC-D09-009 △	Hardening Recommended Practices Document	加固建議實踐文件撰寫	D09.1	DOC	R3	ID13 (Tier 2 format example)	H

D10 — Project Engineering (7 candidates; +1 in R5) ★◆ RE-SCOPED to post-acceptance + decommissioning

8 candidates migrated to D14. 1 new candidate added. See `phase1-revision-r2.md` for migration table.

SC ID	Skill Name (EN)	技能名稱 (ZH)	Subdomain	Type	Lifecycle	Source	Confidence
SC-D10-001	Requirements Traceability Matrix Management	需求追溯矩陣管理	D10.1	DOC	R1,R2	PRAC	M

SC ID	Skill Name (EN)	技能名稱 (ZH)	Subdomain	Type	Lifecycle	Source	Confidence
SC-D10-002	Change Request Evaluation and Impact Analysis	變更申請評估與影響分析	D10.2	ANA	R1-R4	ID01 §6.5.2.3; §7.8.7	H
SC-D10-003	Management of Change (MOC) Execution	變更管理執行	D10.2	MGT	R1-R4	ID01 §7.8.7.1; ID03 §5.7.1	H
SC-D10-004	Technical Clarification Meeting Facilitation	技術澄清會議主持	D10.3	MGT	R1-R3	PRAC	M
SC-D10-005	Contract Technical Scope Tracking	合約技術範圍追蹤	D10.4	MGT	R1-R4	PRAC	H
SC-D10-006 ◆	System Decommissioning Execution	系統除役執行	D10.5	OPS	R5	ID01 §6.5.3	H
SC-D10-007 △	Permit to Work (PtW) Process Management	工作許可流程管理	D10.2	MGT	R3,R4	ID04 §12.0 (Tier 2)	M

D14 — Pre-Gate Engineering (18 candidates; +2 in R5) ★ NEW DOMAIN

10 candidates migrated from former D10 + 6 new candidates.

SC ID	Skill Name (EN)	技能名稱 (ZH)	Subdomain	Type	Lifecycle	Source	Confidence
SC-D14-001	Requirements Specification Development	需求規格書撰寫	D14.1	DOC	Pre-R0	ID03 §5.1.1; Table 2 Doc 0.01	H
SC-D14-002 ◆	Stakeholder Analysis	利害關係人分析	D14.5	ANA	Pre-R0	PRAC	M
SC-D14-003	Technical Feasibility Assessment	技術可行性評估	D14.3	ANA	Pre-R0	PRAC	H
SC-D14-004	Technical Risk Matrix Development	技術風險矩陣建立	D14.6	ANA	Pre-R0	PRAC	M

SC ID	Skill Name (EN)	技能名稱 (ZH)	Subdomain	Type	Lifecycle	Source	Confidence
SC-D14-005	CBOM Development	CBOM 編制	D14.4	ENG	Pre-R0	PRAC	H
SC-D14-006	Labor Hour Estimation	工時估算	D14.4	ANA	Pre-R0	PRAC	H
SC-D14-007	RFI Response Preparation	RFI 回覆準備	D14.1	DOC	Pre-R0	PRAC	M
SC-D14-008	Technical Proposal Writing	技術提案撰寫	D14.3	DOC	Pre-R0	PRAC	H
SC-D14-009	POC Planning and Execution	POC 規劃與執行	D14.3	MGT	Pre-R0	PRAC	M
SC-D14-010	Tender Security Requirements Definition	投標安全需求定義	D14.6	DOC	Pre-R0	ID01 §6.5.1.1.1; ID03 Table 1 SSA-PRO-001	H
SC-D14-011	Site Survey and Constraint Documentation	現場勘查與限制條件文件	D14.2	ANA	Pre-R0	PRAC	H
SC-D14-012	Existing Infrastructure Inventory	既有基礎設施清冊	D14.2	DOC	Pre-R0	PRAC	H
SC-D14-013	Concept Zone/Conduit Architecture	概念 Zone/Conduit 架構	D14.3	DES	Pre-R0	PRAC; ID01 §6.5.1.1	M
SC-D14-014	Preliminary Security Classification	初步安全分類	D14.6	ANA	Pre-R0	ID03 §5.4.1; ID01 §6.5.1.2	H
SC-D14-015	Gate 0 Decision Package Assembly	Gate 0 決策包組裝	D14.6	DOC	Pre-R0	ID01 §6.5.1.1.3; ID03 §5.5.2	H
SC-D14-016	Cost Risk Contingency Analysis	成本風險餘裕分析	D14.4	ANA	Pre-R0	PRAC	M

SC ID	Skill Name (EN)	技能名稱 (ZH)	Subdomain	Type	Lifecycle	Source	Confidence
SC-D14-017 △	NDA and Confidentiality Agreement Management	NDA 與保密協議管理	D14.1	MGT	Pre-R0	ID25 (Tier 3 template)	H
SC-D14-018 ▲	Pre-Gate 0 Requirement Clarification & Feasibility Input	Pre-Gate 0 需求釐清與可行性輸入	D14.3	ANA	Pre-R0	GOV-SD (5 Gate 0 inputs; non-binding boundary)	H

D11 — Engineering Governance (21 candidates; +9 in R5, SC-D11-002 superseded by SC-D11-017)

SC ID	Skill Name (EN)	技能名稱 (ZH)	Subdomain	Type	Lifecycle	Source	Confidence
SC-D11-001	Design Review Checklist Development	設計審查清單建立	D11.1	GOV	R2	PRAC	H
~~SC-D11-002~~	~~Stage-Gate Review Facilitation~~	~~階段閘門審查主持~~	~~D11.1~~	—	—	—	—
SC-D11-003	Critical Security Design Review	關鍵安全設計審查	D11.1	VER	R2	ID01 §7.3.2	H
SC-D11-004	Engineering SOP Development	工程 SOP 制定	D11.2	GOV	R0-R5	PRAC	H
SC-D11-005	Process Efficiency Analysis	流程效率分析	D11.2	ANA	R4	PRAC	L
SC-D11-006	Quality Plan Development	品質計畫撰寫	D11.3	DOC	R1	PRAC	H
SC-D11-007	Nonconformance Management	不合格項管理	D11.3	MGT	R3,R4	ID01 §6.6.4	M
SC-D11-008	Lessons Learned Management	經驗學習管理	D11.4	MGT	R4,R5	PRAC	M

SC ID	Skill Name (EN)	技能名稱 (ZH)	Subdomain	Type	Lifecycle	Source	Confidence
SC-D11-009	Technical Knowledge Base Development	技術知識庫建置	D11.4	ENG	R4	PRAC	L
SC-D11-010	Internal Design Standards Maintenance	內部設計標準維護	D11.5	GOV	R0-R5	PRAC	M
SC-D11-011	Engineering Competency Framework Development	工程能力框架建立	D11.6★◆	GOV	R0	ID01 §6.3; ID03 §5.3.3	H
SC-D11-012	Engineering Training Program Management	工程培訓計畫管理	D11.6★◆	MGT	R0-R5	ID01 §6.3; ID03 §5.3.3	H
SC-D11-013 △	Information Asset Classification and Grading	資訊資產分類分級	D11.3	GOV	R0,R4	ID23 (Tier 3 procedure)	H
SC-D11-014 △	Personnel Security Qualification Management	人員安全資格管理	D11.6	MGT	R0–R5	ID21 (Tier 3); ID06 §5.4	H
SC-D11-015 △	Procurement Security Requirements Integration	採購安全需求整合	D11.2	GOV	R0,R1	ID22 (Tier 3 procedure)	H
SC-D11-016 △	SI/SM Project Security Management Plan Development	SI/SM 專案安全管理計畫撰寫	D11.2	DOC	R0	ID06 (Tier 2; 45+ pages)	H
SC-D11-017 ▲	Gate Review Governance & Blocking Condition Verification	階段審查治理與阻斷條件驗證	D11.1	GOV	R0–R3	GOV-SD (4-gate blocking conditions; 12-item Gate 3 checklist)	H
SC-D11-018 ▲	Design Quality & Traceability Verification	設計品質與追溯性驗證	D11.3	VER	R1,R3	GOV-SD (Design QA role; traceability matrix; document register)	H

SC ID	Skill Name (EN)	技能名稱 (ZH)	Subdomain	Type	Lifecycle	Source	Confidence
SC-D11-019 ▲	Standards Ownership & Exception Arbitration	標準歸屬與例外裁定	D11.2	GOV	R0–R5	GOV-SDP (standards ownership; exception ruling; L1-L2-L3 escalation)	H
SC-D11-020 ▲	Design Change Impact Analysis & SL Recertification	設計變更影響分析與SL重認證	D11.2	ANA	R2	GOV-SD (Gate 2 triggers; SL Decision Record update)	H
SC-D11-021 ▲	Role KPI Evidence Collection & Scoring	角色KPI證據收集與評分	D11.6	MGT	R0–R5	GOV-SDP (SMART KPIs; RCW/AF/PVF formula; bonus mechanism)	M

D12 — Energy Data Platform (8 candidates)

SC ID	Skill Name (EN)	技能名稱 (ZH)	Subdomain	Type	Lifecycle	Source	Confidence
SC-D12-001	Data Acquisition Architecture Design	資料採集架構設計	D12.1	DES	R2	PRAC	M
SC-D12-002	Protocol Parser Development	協定解析器開發	D12.1	ENG	R2,R3	PRAC	M
SC-D12-003	Time-Series Database Selection and Configuration	時序資料庫選型與配置	D12.2	ENG	R2	PRAC	M
SC-D12-004	Data Retention Policy Design	資料保留策略設計	D12.2	DES	R2	PRAC	M
SC-D12-005	Load Forecasting Model Development	負載預測模型開發	D12.3	ENG	R2,R3	PRAC	M
SC-D12-006	Monitoring Dashboard Design	監控儀表板設計	D12.4	DES	R3	PRAC	M

SC ID	Skill Name (EN)	技能名稱 (ZH)	Subdomain	Type	Lifecycle	Source	Confidence
SC-D12-007	Data Dictionary Development	資料字典建立	D12.5	DOC	R2	PRAC	M
SC-D12-008	Data Access Policy Design	資料存取政策設計	D12.5	GOV	R2	PRAC	M

D13 — Engineering Automation (6 candidates)

SC ID	Skill Name (EN)	技能名稱 (ZH)	Subdomain	Type	Lifecycle	Source	Confidence
SC-D13-001	Automated Calculation Script Development	自動化計算腳本開發	D13.1	ENG	R2,R3	PRAC	M
SC-D13-002	Document Generator Development	文件產生器開發	D13.1	ENG	R3	PRAC	M
SC-D13-003	AI-Assisted Design Review	AI 輔助設計審查	D13.2	ENG	R2	PRAC	L
SC-D13-004	Automated Skill Invocation	自動化技能調用	D13.2	ENG	R2-R4	PRAC	L
SC-D13-005	Automated Test Pipeline Development	自動化測試管線開發	D13.3	ENG	R3	PRAC	M
SC-D13-006	Review Workflow Automation	審查工作流程自動化	D13.4	ENG	R2-R4	PRAC	L

Inventory Statistics (R5)

Domain	R3 Count	R5 Count	From ID01-03	From ID04-25/GOV	From PRAC	H/H+ Conf.	M Conf.	L Conf.
D01	28	36	24	8 △▲	4	31	5	0
D02	12	12	4	0	8	4	7	1
D03	10	10	0	0	10	2	7	1
D04	6	6	0	0	6	3	3	0
D05	14	14	0	0	14	5	9	0

Domain	R3 Count	R5 Count	From ID01-03	From ID04-25/GOV	From PRAC	H/H+ Conf.	M Conf.	L Conf.
D06	6	6	0	0	6	4	2	0
D07	7	7	0	0	7	1	6	0
D08 ◆	11	13	8	2 △	3	11	1	1
D09	8	9	3	1 △	5	6	3	0
D10 ★◆	6	7	3	1 △	3	3	3	1
D11	12	21	5	9 △▲	7	14	4	3
D12	8	8	0	0	8	0	8	0
D13	6	6	0	0	6	0	3	3
D14 ★	16	18	4	2 △▲	12	10	5	3
Total	**150**	**173**	**51**	**23**	**99**	**94**	**66**	**13**

R5 Normalization notes:

- R3 post-normalization: 149 (SC-D09-007 merged into SC-D11-004)
- R5 post-normalization: **171** (SC-D09-007 merge + SC-D11-002 superseded by SC-D11-017)
- H+ confidence subset: 19 candidates with both Tier 2 exemplar and Tier 1 governance confirmation (see R5 analysis for details)
- Source tier breakdown: 51 from Tier 2 originals (ID01-03), 15 from Tier 2 exemplars (ID04-14), 5 from Tier 3 org procedures (ID21-25), 8 from Tier 1 governance (GOV-SD/SDP), 99 from PRAC